

WESTFIELD-BARNES, MA, USA

WMO: 744915

Lat: 42.158N

Lon: 72.716W

Elev: 271

StdP: 14.55

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 14775

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.7	6.1	-11.1	3.0	5.5	-7.2	3.8	9.5	26.4	22.8	24.2	25.2	4.7	10	0.507

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.8	91.2	72.4	88.4	71.2	85.3	69.5	75.8	85.5	74.4	82.9	72.9	80.7	10.0	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.1	124.1	79.1	72.0	119.7	78.0	70.4	113.2	76.9	39.4	85.7	37.9	83.0	36.6	80.7	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.0	18.9	16.9	-7.3	96.4	5.7	3.0	-11.4	98.6	-14.7	100.4	-17.9	102.1	-22.1	104.3		
			-8.0	78.7	5.5	1.2	-11.9	79.6	-15.1	80.3	-18.2	81.0	-22.1	81.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.7	24.8	27.7	36.0	48.4	59.1	67.3	72.9	71.1	63.8	51.5	40.9	31.2		
	DBStd	18.47	10.58	9.38	9.08	8.62	7.91	6.61	5.51	5.48	7.09	8.11	8.41	9.09		
	HDD50	2950	781	625	445	130	11	0	0	0	1	80	293	584		
	HDD65	6283	1245	1044	900	505	222	51	6	12	106	423	722	1047		
	CDD50	2832	1	1	10	81	291	519	709	653	416	127	21	3		
	CDD65	691	0	0	0	6	38	120	250	200	71	6	0	0		
	CDH74	6607	0	1	5	109	514	1193	2405	1739	596	45	0	0		
	CDH80	2290	0	0	0	39	177	405	931	568	164	6	0	0		
	WSAvg	6.5	7.2	7.6	8.2	7.8	6.6	5.8	5.4	4.9	5.2	6.0	6.6	6.9		
	PrecAvg	46.3	3.1	2.8	3.9	3.6	3.7	4.3	4.1	4.5	4.4	4.3	3.4	3.9		
Precipitation	PrecMax	65.3	6.6	9.3	7.0	7.5	7.0	10.1	10.0	10.0	10.4	13.8	5.8	6.9		
	PrecMin	32.0	1.3	1.4	0.7	1.0	1.1	1.7	2.0	1.2	1.2	0.5	0.6	0.8		
	PrecStd	8.2	1.2	1.5	1.6	1.6	1.5	2.3	2.2	2.3	2.8	2.8	1.2	1.4		
	PrecStd	8.2	1.2	1.5	1.6	1.6	1.5	2.3	2.2	2.3	2.8	2.8	1.2	1.4		
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.2	59.2	70.7	85.9	90.4	93.1	95.9	93.1	90.4	79.4	68.4	61.7		
		MCWB	52.4	50.6	54.3	63.3	67.5	72.9	75.6	74.3	73.0	66.6	58.9	56.0		
	2%	DB	48.5	50.9	61.7	75.3	85.0	88.5	91.5	89.1	84.4	73.4	63.6	54.1		
		MCWB	43.0	43.5	48.9	56.7	66.2	70.8	72.9	73.0	69.6	63.1	56.1	49.0		
	5%	DB	42.9	45.2	55.2	69.7	80.0	84.4	88.8	85.8	80.5	69.8	59.3	49.0		
		MCWB	38.1	38.1	44.9	54.6	63.1	68.7	71.9	70.5	67.4	62.1	52.6	44.2		
	10%	DB	38.6	40.0	50.0	64.3	74.6	81.2	85.1	82.8	76.9	65.5	54.9	44.6		
		MCWB	34.3	34.5	42.8	51.6	60.7	66.9	70.1	69.4	65.8	58.1	48.7	39.7		
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.1	54.4	57.6	65.3	72.0	76.2	78.4	77.4	75.3	70.2	63.1	56.7		
		MCDB	55.4	56.8	66.0	80.3	85.3	87.4	90.3	87.7	83.9	74.7	65.9	58.8		
	2%	WB	43.8	44.3	51.6	61.6	68.9	74.0	76.1	75.4	72.8	66.7	58.2	50.3		
		MCDB	47.4	50.4	58.0	72.6	79.7	83.8	86.3	83.7	79.9	70.9	61.6	52.9		
	5%	WB	38.6	38.7	46.9	57.5	66.4	72.0	74.5	73.9	70.9	63.2	53.8	44.6		
		MCDB	42.3	43.8	53.8	64.6	75.5	80.6	83.4	81.3	77.3	68.0	57.8	48.3		
	10%	WB	35.0	35.7	43.1	53.4	63.5	69.9	73.0	72.3	68.7	59.3	49.7	40.2		
		MCDB	38.1	39.3	49.5	61.7	70.7	77.0	81.0	79.1	74.1	64.5	54.6	43.9		
Mean Daily Temperature Range	5% DB	MDBR	16.9	18.1	18.8	22.3	22.7	21.6	21.8	21.5	22.2	20.7	18.6	15.7		
		MCDBR	20.2	23.7	28.1	32.7	30.8	26.8	25.6	24.5	25.8	25.5	24.6	21.2		
		MCWBR	16.2	16.8	17.3	16.8	14.4	12.4	10.0	10.2	12.7	15.2	17.3	17.6		
	5% WB	MCDBR	19.3	21.8	24.8	27.0	25.5	23.0	21.3	19.7	20.1	21.1	20.8	19.7		
		MCWBR	16.3	16.3	17.2	16.0	13.4	12.0	9.6	9.4	11.2	15.0	16.9	17.7		
Clear-Sky Solar Irradiance	taub		0.294	0.309	0.334	0.370	0.408	0.455	0.473	0.442	0.382	0.344	0.312	0.295		
	taud		2.427	2.394	2.418	2.365	2.295	2.194	2.151	2.226	2.371	2.491	2.536	2.517		
	Ebn at Noon		271	284	290	285	276	262	255	260	269	269	264	262		
	Edh at Noon		24	30	33	37	41	45	47	43	34	27	22	21		
All-Sky Solar Radiation	RadAvg		559	831	1157	1478	1648	1785	1858	1658	1357	881	613	456		
	RadStd		43	75	85	159	176	172	112	89	105	89	54	47		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.15	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (82 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+182	+90

Nomenclature: See separate page